

Registration No.:

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Total Number of Pages: 02

Integrated Dual Degree (B.Tech and M.Tech)
REL4D003/ REE4D003

4th Semester Regular/Back Examination: 2023-24
SUBJECT: Electrical and Electronics Measurement
BRANCH(S): ELECTRICAL, EEE, EE

Time: 3 Hours
Max Marks: 100
Q.Code: P589

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions: (2 x 10)

- What are the different methods used for measurement of medium and high resistance.
- The measured value of a capacitor is 205.3 μ F, whereas its true value is 201.4 μ F. Find the value of relative error.
- Explain the basic principle of operation of a constant resistance d.c. potentiometer.
- Explain the difference between gravity control and spring control.
- What is the Lissajous pattern with equal voltages of equal frequency and phase shift by 90°?
- Explain the phenomena of Creeping. If an energy meter disc makes 10 revolutions in 100 seconds when a load of 360 W is connected to it, determine the meter constant revolution/KWh.
- A moving coil of a meter has 100 turns, and a length and depth of 10mm and 20mm, respectively. It is positioned in a uniform radial flux density of 200mT. The coil carries a current of 50mA. Find the torque on the coil.
- Define Logarithmic decrement in Galvanometer.
- What do you understand by burden in instrument transformer?
- In a digital voltmeter the oscillator frequency is 400 kHz, the ramp voltage falls from 8V to 0V in 20 msec. Find the number of pulses counted by the counter.

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- A coil of a ballistic galvanometer has 115 turns of mean area 25 $\times$ 40 mm². The flux density in the air gap is 0.12 wb/m² and the moment of inertia is 0.5 $\times$ 10⁻⁶ Kg-m². The stiffness is 45 $\times$ 10⁻⁶ Nm/rad. What current must be passed to give a deflection of 1000° and what resistance must be added in series with the movement to give critical damping?
- Explain how voltage and Current are measured with the help of a CRO.
- Explain working principle of mechanical resonance type frequency meter.

- d) A 4 terminal resistor of approximately 50 μohm resistance was measured by means of a Kelvin bridge having the following component:
Standard resistor = 100.03 μohm , Inner ratio arms = 100.31 μohm and 200 μohm :
outer ratio arms = 100.24 ohm and 200 ohm , resistance of link connecting the standard and the unknown resistance = 700 μohm . Calculate the unknown resistance.
- e) The exciting current of a ring core CT of ratio 1000/5 A. When operating at full primary current and with a secondary burden of non inductive resistance of 1 ohm is 1A at a p.f. of 0.4. (I) Calculate the phase displacement between primary and secondary current. (II) The ratio error at full load, assuming that there has been no compensation.
- f) If the vertical amplifier of an oscilloscope has a bandwidth of 15MHz. What is the fastest rise time that an input may have to be displayed without distortion?
- g) Draw the block diagram of digital voltmeter.
- h) Explain different types of errors in measuring instrument.
- i) The inductive reactance of the pressure coil circuit of a dynamometer wattmeter is 0.4% of its resistance at normal frequency 50 Hz and the capacitance is negligible, Find the percentage error and correction factor due to the reactance for load at 0.707 p.f. lagging.
- j) In an indicating PMMC type ammeter the current through the coil for full scale deflection is 5mA. Coil resistance is 5 ohm and swamping resistance is 4 ohm . The meter is to be used for measuring current up to 1 ampere. Calculate the value of shunt resistance required. What arrangement is required to be made to compensate to temperature change which may occur? Derive necessary expression.
- k) Describe the construction and working principle of a single-phase energy meter.
- l) Explain the balanced condition of Anderson's bridge with appropriate phasor diagram, also state the advantage and disadvantage of the bridge.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3 a) Describe the construction details of an electrodynamic type wattmeter. Derive the expression for torque when the instrument is used on AC. (8x2)
b) Explain the working principle of Spectrum Analyzer.
- Q4 a) Explain the construction and principle of working of a Linear Variable Differential Transformer (LVDT). Explain how the magnitude and direction of the displacement of core of an LVDT detected. (8x2)
b) Explain Measurement of Mutual Inductance by Felici's Method.
- Q5 Draw the equivalent circuit and phasor diagram of a Current transformer. Derive the expressions for its ratio and phase angle errors. Describe the assumptions made for deviation of errors. (16)
- Q6 Write short notes on (8x2)
(a) Hall Effect Transducer
(b) Power Factor Meters

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Course: B.Tech

Sub_Code: REE4D003/REL4D003

4th Semester Regular / Back Examination: 2022-23

SUBJECT: Electrical and Electronics Measurement

BRANCH(S): EEE, ELECTRICAL

Time : 3 Hour

Max Marks : 100

Q.Code : M606

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions:

- State the difference between accuracy and precession.
- What are the disadvantages of Hay's bridge in context to Q-factor?
- Define Logarithmic decrement in Galvanometer.
- A moving coil of a meter has 100 turns, and a length and depth of 10mm and 20mm, respectively. It is positioned in a uniform radial flux density of 200mT. The coil carries a current of 50mA. What is the torque on the coil?
- What is the principle of operation of an LVDT?
- What is the Lissajous pattern with equal voltages of equal frequency and phase shift by 90°?
- Explain how Kelvin double bridge is special from Wheatstone bridge.
- What do you understand by burden in instrument transformer?
- What is electronic voltmeter and what are its advantages?
- Define distortion factor of a signal.

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 × 8)

- Describe the construction and principle of operation of single phase energy meter.
- A 4-terminal resistor of approximately 50 μ ohm resistance was measured by means of a Kelvin bridge having the following component: Standard resistor = 100.03 μ ohm, Inner ratio arms = 100.31 μ ohm and 200 μ ohm: outer ratio arms = 100.24 ohm and 200 ohm, resistance of link connecting the standard and the unknown resistance = 700 μ ohm. Calculate the unknown resistance.
- The meter constant of a 230V, 10A watt-hour meter is 1800 revolutions per KWh. The meter is tested at half load and rated voltage and unit power factor. The meter is found to make 80 revolutions in 138s. Determine the meter error at half load.

- d) Discuss Systematic errors giving suitable examples. Explain the measures taken to minimize these errors.
- e) Explain difference between gravity control and spring control.
- f) Explain the Measurement of Mutual Inductance by Felici's method.
- g) Derive the expression for the value of capacitance to compensate the frequency error in moving iron type instrument.
- h) Explain the construction and principle of working of an L.V.D.T. Explain how the magnitude and direction of the displacement of core of an L.V.D.T. is detected.
- i) If the vertical amplifier of an oscilloscope has a bandwidth of 15MHz. What is the fastest rise time that an input may have to be displayed without distortion?
- j) The deflecting torque of an ammeter is directly proportional to the current through it. If a current of 10A deflects the pointer by 90° , find the value of current for a deflection of 60° when the instrument is (a) spring controlled and (b) gravity controlled.
- k) What are the problems associated with a.c. potentiometer? Describe the working of Gall-Tinsley Potentiometer.
- l) Draw the circuit diagram and phasor diagram of Maxwell's bridge.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- | | | |
|-----------|---|-------------|
| Q3 | Draw the equivalent circuit and phasor diagram of a current transformer. Derive the expressions for its ratio and phase angle errors. | (16) |
| Q4 | Describe the Construction, Theory and Principle of operation and errors of Electro-Dynamometer type wattmeter. | (16) |
| Q5 | Write short notes on
(a) Vibrating reed type Frequency Meter
(b) Digital storage Oscilloscope | (16) |
| Q6 | Derive the expressions for the deflection of D' Arsonval type galvanometer when the meter is operating under (i) under-damped condition and (ii) critically damped condition. | (16) |

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B. TECH

PEE3I104, PEI3I102, PEL3I104

3rd Semester Back Examination: 2022-23
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH(S): ELECTRICAL, EIE, EEE

Time : 3 Hour

Max Marks : 100

Q.Code : L544

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions :

(2 x 10)

- Why the secondary of a CT is never left open circuited?
- Give at least two most common methods for measurement of low resistance.
- What are meter phase error and creep error, and how it can be removed from the instrument?
- An ammeter of 0-25 A range has a guaranteed accuracy of 1% of full scale reading. The current measured is 5 A. What is the value of limiting error?-
- A Galvano meter has a resistance of 50 Ω and a range of (0-50 μ A). What value of shunt is to be used so that it can measure current up to 5mA?
- What is a Q-meter?
- Briefly differentiate between Accuracy and Precision of measurement.
- Two resistors $R_1 = 36 \Omega \pm 5\%$ and $R_2 = 75 \Omega \pm 5\%$ are connected in series. What is the total value of resistance?
- What is loss of charge method?
- Why in a D'Arsonval galvanometer, an iron core is usually used between the permanent magnet pole faces?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Explain the construction and working of Single-phase Energy meter with brief circuit diagram?
- Draw the phasor diagram of a CT & discuss the effect of variation of power factor of the secondary burden upon the performance.
- Derive the torque equation for a dynamometer type wattmeter. Comment upon the shape of scale if spring control is used.
- Describe the construction and working of a polar type potentiometer?
- Explain about the principle of operation of Ballistic Galvanometer and find the derivation for measurement of flux density through this instrument?
- Write short notes on rectifier type AC voltmeters?
- Derive the torque expression of an Electro dynamo meter type instrument?
- Write short notes on flux meter.

- i) Derive the expression for external resistance required for critical damping of a galvanometer.
- j) Find the expression for Self Inductance using Anderson's bridge method?
- k) Describe a true r.m.s reading voltmeter with neat sketches.
- l) Derive the equation of balance of a Schering Bridge. Draw the phasor diagram under null conditions and explain how loss angle of capacitor can be calculated.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** What are the different problems associated with measurement of low resistances and how these are eliminated using Kelvin' Double Bridge? Draw the circuit of a Kelvin' Double Bridge and derive the condition for balance. **(16)**
- The four arms of a wheatstone bridge are as follows:
 $AB=100\Omega$; $BC=1000\Omega$, $CD=4000\Omega$, $DA=400\Omega$. The galvanometer has a resistance of 100Ω and, a sensitivity of $100\text{mm}/\mu\text{A}$ and is connected across AC. A source of 4 V d.c. is connected across BD. Calculate the current through the galvanometer and its deflection if the resistance of arm DA is changed from 400Ω to 401Ω .
- Q4** Discuss the common sources of error in AC bridge. How are they eliminated? **(16)**
- Q5** Derive the expressions for ratio and phase angle errors for current transformer and discuss the effect of secondary burden on it. **(16)**
- Q6** Draw block diagram of an oscilloscope and explain its operation. Explain the methods of measurement of frequency, phase angle and time delay using oscilloscope. **(16)**

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B.Tech
REE4D003 / REI4C001 / REL4D003

4th Semester Regular/ Back Examination: 2021-22
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH(S): EEE, AEIE, ELECTRICAL

Time : 3 Hour

Max Marks : 100

Q.Code : J776

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions :

(2 × 10)

- State the difference between accuracy and precession.
- What is creep error in a single phase induction type energy meter.
- A PMMC meter has an internal resistance 200Ω and the current required for its full scale deflection is $50\mu A$. What will be the maximum measuring voltage of the meter.
- Discuss the limitations of Wheatstone bridge.
- What is the difference between gravity control and spring control?
- State The advantage of Hay's bridge over Maxwell's Inductance-Capacitance bridge is
- Define standard deviation and variance in the context of normal frequency distribution of data.
- Explain the basic principle of operation of a d.c potentiometer.
- Describe the IEEE Standards.
- Write the applications of a CRO.

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 × 8)

- Discuss the measurement of capacitance by Schering Bridge.
- Define systematic errors giving suitable examples. Explain the measures taken to minimize these errors.
- The meter constant of a 230V, 10A watt-hour meter is 1800 revolutions per KWh. The meter is tested at half load and rated voltage and unity power factor. The meter is found to make 80 revolutions in 138s. Determine the meter error at half load.
- The exciting current of a ring core current transformer of ratio 1000/5A when operating at full primary current and with a secondary burden of non-inductive resistance of 1Ω is 1A at a power factor of 0.4. Calculate (i) the phase displacement between primary and secondary winding currents (ii) the ratio error at full load, assuming that there has been no compensation.
- Derive the expression of the value of capacitance to compensate the frequency error in moving iron type instrument?

- f) Draw the block diagram of digital voltmeter.
- g) The coil of a 300V moving iron voltmeter has a resistance of 500Ω and an inductance of $0.8H$. The instrument reads correctly at $50Hz$ a.c supply and takes $100mA$ at full scale deflection. What is the percentage error in the instrument reading when it is connected to $200V$ d.c supply.
- h) Explain the construction and principle of working of a Linear Variable Differential Transformer (LVDT). Explain how the magnitude and direction of the displacement of core of an LVDT detected?
- i) A 4 terminal resistor of approximately $50\mu\Omega$ resistance was measured by means of a Kelvin bridge having the following component: Standard resistor = $100.03\mu\Omega$, Inner ratio arms = $100.31\mu\Omega$ and $200\mu\Omega$; outer ratio arms = 100.24Ω and 200Ω , resistance of link connecting the standard and the unknown resistance = $700\mu\Omega$. Calculate the unknown resistance.
- j) Explain the characteristics of instruments and measurement system.
- k) What is Phantom loading? Explain with an example how is it more advantageous than testing with direct loading.
- l) Describe how frequency and phase angle measurement can be done with the use of a CRO.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** Describe constructional details and principle of operation of a d' Arsonval galvanometer. Derive the expression for steady state deflection. If the instrument is provided with a mirror and a light source is placed $1m$ away from the galvanometer, find the expression for the deflection of light spot. **(16)**
- Q4** Define the working of PMMC instrument. Derive the equation for deflection if the instrument is spring controlled. Describe methods of damping used in these instrument. **(16)**
- Q5** Draw the equivalent circuit and phasor diagram of a potential transformer. Derive the expressions for its ratio and phase angle errors. Describe the assumptions made for deviation of errors. **(16)**
- Q6** Write short notes on: **(16)**
 (a) Measurement of Mutual Inductance by Felici's Method
 (b) Wagner Earthing Device

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B.Tech
PEE3I104 / PEI3I102 / PEL3I104

3rd Semester Back Examination: 2021-22
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH(S): ELECTRICAL / AEIE, EIE / EEE

Time : 3 Hour

Max Marks: 100

Q.Code: OF655

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right-hand margin indicate marks.

Part-I

Q1 Answer the following questions: (2×10)

- The current flowing in a circuit consists of a DC of 3A and an AC of peak value 2A. The readings of a moving coil ammeter and moving iron ammeter will be?
- In two wattmeter method of measuring 3-phase power, the power factor is 0.5, then one of the watt meters will read?
- Define the term precision with a suitable physical example?
- Which type of damping is most employed in the dynamometer type wattmeter?
- State the term accuracy with a suitable physical example.
- Give the different stages inside the CRT tube?
- Which type of controlling torque is used in a single-phase power factor meter?
- A 0 to 200V voltmeter has a guaranteed accuracy of 1% of full-scale reading. The voltage measured by this instrument is 50V. What is the limiting error?
- The power of a method three-phase, the three-wire balanced system was measured by two wattmeter method. The reading of one of the watt meters was found to be double that of the other. What is the power factor of the system?
- An energy meter having a meter constant of 1200 rev per KWH is found to make 5 revolutions in 75 s. What is the load power?

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6×8)

- Draw the Circuit of the Kelvin double ratio arm bridge and show the effect of lead resistance is eliminated. Draw and explain the megaohm bridge method for high resistance measurement.
- Explain the working principle of the mechanical resonance type frequency meter?
- A linear resistance potentiometer is 100 mm long and is uniformly wound with a wire of total resistance of 10000 ohms. Under normal conditions, the slider is at the center of the potentiometer. Determine the linear displacement when the resistance of the potentiometer as measured by the whetstone bridge is 3700 ohm. If it is possible to measure a minimum value of 5-ohm resistance with the above arrangement. Determine the resolution of the potentiometer in mm.
- Describe the construction and working principle of a single-phase energy meter.

- e) Explain the phenomena of Creeping. If an energy meter disc makes 10 revolutions in 100 seconds when a load of 360w is connected to it, determine the meter constant revolution/Kwh.
- f) The exciting current of a ring core CT of ratio 1000/5 A. When operating at full primary current and with a secondary burden of noninductive resistance of 1ohm is 1A at a p.f of 0.4. (i) Calculate the phase displacement between primary and secondary current. (ii) The ratio error at full load, assuming that there has been no compensation
- g) Describe a method of power measurement for 3 phase 3wire unbalance Load. How can the power factor of a balanced load be determined by the method?
- h) Draw an electrodynamics type frequency meter circuit and explain its working.
- i) Write down the theory and principle of operation of D' Arsonval Galvanometer?
- j) Draw the circuit of Schering Bridge. Derive the null conditions. Draw the phase diagram of Schering Bridge.
- k) Derive the expression for transformer ratio and phase angle of CT?
- l) In the Wheatstone bridge, the value of resistances of various arms is P= 1000ohm, Q=100ohm, R=2005ohms, and S=200ohm. The battery has an emf of 5V and negligible internal resistance. The galvanometer has a current sensitivity of 10mm/microampere and internal resistance of 100ohm. Calculate the deflection of the galvanometer and the sensitivity of the bridge in terms of deflection per unit change in resistance.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** a) Draw the circuit of Anderson Bridge. Derive the null conditions. Draw the phase diagram of Anderson Bridge. (8)
- b) Draw the circuit of the Maxwell-inductance Bridge. Draw the phase diagram of the Maxwell-inductance Bridge. (8)
- Q4** a) Explain the Electrodynamometer wattmeter and derive the torque equation. (8)
- b) The inductive reactance of the pressure coil circuit of a dynamometer wattmeter is 0.4% of its resistance at a normal frequency of 50 Hz and the capacitance is negligible, Find the percentage error and correction factor due to the reactance for load at 0.707 p.f. lagging. (8)
- Q5** a) Derive the general torque equation for the Moving iron instrument. (8)
- b) The inductance of a moving iron meter is given by the following expression: $L=20+10\alpha-2\alpha^2$ micro henry, where α is deflection in radians. The spring constant is 24×10^{-6} Nm/rad. Calculate the value of deflection for a current of 5A. (8)
- Q6** Draw the basic block diagram of an oscilloscope and state the function of each block. Describe the Oscilloscope specifications and performance. (16)

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B.TECH
PEE3I104

3rd Semester Back Examination 2019-20
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH : ELECTRICAL

Max Marks : 100

Time : 3 Hours

Q.CODE : HB683

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) (2 x 10)

- Give one example each for (i) an Absolute Instrument (ii) a Secondary Instrument.
- In a D'Arsonval galvanometer, an iron core is usually used between the permanent magnet pole faces, explain with reasons.
- A moving coil instrument gives full scale deflection of 15mA when the potential difference across its terminals is 300mV . Calculate, the shunt resistance for measuring upto 25 Amp.
- Give two examples of (i) Integrating type of Instruments (ii) Secondary Instruments.
- Give at least two most common methods for measurement of low resistance.
- For 20A, 230V energy meter, the revolution per Kilowatt-hour is 480. If upon test at full load unit power factor the disc makes 40 revolution in 66 seconds, calculate the error in the energy meter.
- Differentiate between sensors and transducers.
- Briefly explain, "standardization" in a slide wire potentiometer.
- A Lissajous pattern on an oscilloscope is stationary and has 5 vertical maximum values and 4 horizontal maximum values. The frequency of the horizontal input is 1200 Hz. What is the frequency of vertical input?
- Give Reasons, the secondary of a CT is never left open circuited.

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- What are the differences between static and dynamic characteristics of instruments?
- Discuss the theory and principle of operation of Electro-Dynamometer type wattmeter.
- Derive the equation of balance of a Schering Bridge. Draw the phasor diagram under null conditions and explain how loss angle of capacitor can be calculated.
- The following readings were obtained during the measurement of a low resistance using a potentiometer. Voltage drop across a 0.1Ω standard resistance is 1.0235V Voltage drop across the low resistance under test=0.4221V Calculate the value of unknown resistance, current and power lost in it.

- e) Briefly explain the calibration and adjustments of a single-phase induction type energy meter.
- f) Derive the equation of balance of a Schering Bridge. Draw the phasor diagram under null conditions and explain how loss angle of capacitor can be calculated.
- g) Discuss the importance of Wagner Earthing Device in AC bridges.
- h) Explain the term standardization of a potentiometer. Describe the procedure of standardization of a d.c. potentiometer.
- i) Explain the operation of LVDT, with help of a diagram.
- j) How the frequency is converted to an analog signal? Explain.
- k) Explain any one bridge circuit for measurement of Inductance.
- l) Discuss the common sources of error in an AC bridge. How are they eliminated?

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** a) Describe the construction and working of PMMC instrument. Derive the equation for deflection if the instrument is spring controlled. **(10)**
- b) State various types of frequency meters and explain the working of a vibrating reed type frequency meter. **(6)**
- Q4** a) What is a megger? Why is it used? Explain the working principle of Megger with relevant diagram. **(10)**
- b) Explain the Kelvin's Double bridge and obtain the balance condition. **(6)**
- Q5** a) Describe the working of a Ballistic Galvanometer and compare it with a D'Arsonval Galvanometer. **(10)**
- b) Discuss the constructional features of a Polar type Potentiometer. **(6)**
- Q6** a) With a neat diagram explain the main parts and working of Cathode Ray Oscilloscope. With a neat schematic, explain the operation of a dual slope analog to digital conversion. **(10)**
- b) Describe a true r.m.s reading voltmeter with neat sketches. **(6)**

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B.Tech
PET4I103

2nd Year Special Examination 2018-19
ELECTRICAL AND ELECTRONICS MEASUREMENT

BRANCH : ECE, ETC

Time : 3 Hours

Max Marks : 100

Q.CODE : G843

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part-I

Q1 Answer the following questions : Short answer type (2 x 10)

- A resistor whose true value is 1,684,465 ohm is measured by an ohmmeter which consistently and repeatedly indicates 1.7 Mega ohms. Identify the type of error.
- Write the operation of a head phone used as a detector in AC bridges.
- How the construction of a D'Arsonval type of galvanometer is different from Ballistic galvanometer?
- Why electro-dynamometer type instruments are called as transfer instruments?
- Comment on the loading effect of a potentiometer when used for measuring unknown voltage of any circuit.
- What is creeping in single phase induction type energy meter and how is it avoided?
- Draw circuit diagram of a peak-reading voltmeter.
- Write basic principle of operation of a current probe used for oscilloscope.
- Define "Harmonic distortion".
- Draw circuit diagram of an average responding voltmeter.

Part-II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- With neat diagram explain the principle of operation, accuracy of measurement of a digital multi-meter.
- Distinguish between gross error, systematic error and random error with examples. Suggest methods for their reduction or elimination.
- Explain the working of attraction type of moving iron instrument with the help of neat diagram.
- Explain the construction and working of a D'Arsonval galvanometer.
- Describe the procedure of standardization of an a.c. potentiometer.
- Describe the construction, principle of working and applications of Hall effect transducers.
- Explain method of protecting galvanometer when used as a detector in Wheatstone bridge.
- Derive an expression of the deflecting torque of an electro-dynamometer type instrument.
- With suitable diagram explain the operation of a Spectrum Analyzer.
- Describe operations of various type of Delay lines circuits used in Oscilloscope.
- Describe the construction and basic principle of operation of any one frequency meter.
- Enumerate the delay time. How do you obtain the delay?

Part III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3

Describe the working of Owen’s bridge. Derive the equations for balance and draw the phasor diagram under conditions of balance.

(16)
- Q4

Draw the equivalent circuit and phasor diagram of a Potential transformer. Derive the expressions for ratio and phase angle errors.

(16)
- Q5

Enumerate the Kelvin’s double bridge measurement with a neat sketch. Derive the necessary expressions to support your answer. Draw the phasor diagram.

(16)
- Q6

Write short notes on :

(16)

a)

Gall-Tinsley potentiometer

b)

Circuit diagram and calibration of Vernier type potentiometer

c)

Q-meter Oscilloscope

d)

Dual ramp type digital voltmeter

Registration No :

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B.Tech
PEE3I104

2nd Year Special Examination 2018-19
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH : ELECTRICAL

Max Marks : 100

Time : 3 Hours

Q.CODE : G399

Answer Question No.1 (Part-1) which is compulsory, any EIGHT from Part-II and any TWO from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) (2 x 10)

- In a portable resistance testing set, write the range of voltage generated for measurement of insulation resistance.
- Write different categories of systematic errors. Give one example of each category of errors.
- How the construction of a D'Arsonval type galvanometer is different from Ballistic galvanometer?
- Comment on the loading effect of a potentiometer when used for measuring unknown voltage of any circuit.
- How can you reverse the meter in a household single phase induction type wattmeter? Justify.
- What is the power consumed by a potentiometer?
- Why do you use a Schmitt trigger circuit in a digital frequency meter?
- What are the advantages of electronics voltmeter comparing to electromechanical type voltmeter?
- What will happen if the time period of sweep voltage applied to X-input of an oscilloscope increases?
- Write mathematical expressions of Harmonic distortion and total harmonic distortion.

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Describe the construction and working of PMMC instrument. Derive the equation for deflection if the instrument is spring controlled.
- Distinguish between gross error, systematic error and random error with examples. Suggest methods for their reduction or elimination.
- Explain the working of attraction type of moving iron instrument with the help of neat diagram.
- Explain the construction and working of a D'Arsonval galvanometer.
- Describe the procedure of standardization of a d.c. potentiometer.
- Describe the construction, principle of working and applications of Hall effect transducers.
- Explain the measurement of self-inductance using voltmeter-ammeter method.
- Draw a basic circuit of a digital frequency meter using various sections. Explain the function of each section.
- Draw the block diagram of electronic voltmeter and explain the principle of operation.
- With neat diagram explain the principle of operation, accuracy of measurement of a digital multi-meter.
- Draw the circuit diagram and explain the operation of single-phase Electro-dynamometer type power factor meter.
- Describe operations of various type of delay lines circuits used in Oscilloscope.

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

Q3 Describe the working of Schering bridge. Derive the equations for balance and draw the phasor diagram under conditions of balance. **(16)**

Q4 Draw the equivalent circuit and phasor diagram of a current transformer. Derive the expressions for ratio and phase angle errors. **(16)**

Q5 a) Explain the function and working of Wagner Earth Devices. **(8)**

b) Draw the block diagram of Dual ramp type digital voltmeter and explain its operation. **(8)**

Q6 Write short notes on :

a) Gall-Tinsley potentiometer **(8)**

b) Q-meter Oscilloscope **(8)**

Registration No :

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Total Number of Pages : 01

B.Tech
PCEE4204

4th Semester Back Examination 2018-19
ELECTRICAL AND ELECTRONICS MEASUREMENT

BRANCH: EEE, ELECTRICAL

Time : 3 Hours

Max Marks : 70

Q. CODE : F1007

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right-hand margin indicate marks.

- Q1** Answer the following questions : (2 x 10)
- a) Explain the term "static stiffness" and "static compliance".
 - b) Distinguish between fundamental and derived units.
 - c) In calculating voltage drop, a current of 4.37 A is recorded in a resistance of 31.27 ohm. Calculate the voltage drop across the resistor to the appropriate number of significant figures
 - d) Explain the use of instrument transformers.
 - e) Why is Wagner ground connection useful?
 - f) Mention two applications of Wien Bridge.
 - g) Describe the term "standardization", of a d.c. potentiometer. How is the standardization done for an a.c. potentiometer?
 - h) Compare ballistic galvanometer and vibration galvanometer.
 - i) What is aquadag and why is it required?
 - j) What is a volt-ratio box?
- Q2** a) Describe about different types of errors in measurement. (5)
b) Explain about all the static characteristics of measurement. (5)
- Q3** a) Illustrate the measurement of insulation resistance of a cable using two-wire system. (5)
b) Compare the operation of Drysdale-Tinsley & Gall-Tinsley Potentiometer. (5)
- Q4** a) Explain briefly the effect of galvanometer resistance on damping. (5)
b) Derive the equations for balance for the bridge for measurement of low resistance. (5)
- Q5** a) How is true RMS responding voltmeter is advantageous? Explain it's working. (5)
b) Explain about working of CRT with detailed diagram. (5)
- Q6** Describe the principle of working of a moving iron instrument. How are they classified. Show that this type of instrument can be used for both d.c. and a.c. measurements. (10)
- Q7** a) Derive the expression of gauge factor for a metallic wire strain gauge. (5)
b) A resistance, wire strain gauge with a gauge factor of 2 is bonded to a steel structural member subjected to a stress of 100 MN/m². The modulus of elasticity of steel is 200GN/m². Calculate the percentage change in the value of the gauge resistance due to the applied stress. (5)
- Q8** Write short answer on any TWO : (5 x 2)
- a) Ramp type DVM
 - b) Construction and working of a PMMC instrument
 - c) Anderson bridge

Registration No :

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Total Number of Pages : 02

B.Tech
PET4I103

4th Semester Regular / Back Examination 2018-19
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH : ECE, ETC
Time : 3 Hours
Max Marks : 100
Q.CODE : F835

Answer Question No.1 (Part-1) which is compulsory, any eight from Part-II and any two from Part-III.

The figures in the right hand margin indicate marks.

Part- I

Q1 Only Short Answer Type Questions (Answer All-10) (2 x 10)

- How the IEEE standard is different from other standards?
- Draw a graph to distinguish between low accuracy & high precision
- Name two null detectors for ac bridges.
- Name the bridge circuit that measures unknown inductance in terms of a known capacitance. Draw the circuit.
- How is Wagner ground connection useful?
- Write down important applications of D.C POT (at least 4).
- Compare ballistic galvanometer and vibration galvanometer.
- Distinguish between CT and PT.
- Why is the FET input D.C voltmeter is required?
- How are digital voltmeters broadly classified?

Part- II

Q2 Only Focused-Short Answer Type Questions- (Answer Any Eight out of Twelve) (6 x 8)

- Explain all the static characteristics of measurement.
- Classify different standards based on their use.
- Develop the equation of balance for the bridge used for measurement of inductance for high Q-coil.
- If $Z_1 = 450 \Omega$, $Z_2 = (300-j600) \Omega$, $Z_3 = (200+j100) \Omega$, then find out the value of Z_4 , so that, the bridge is balanced.
- Deduce the equation of balance for the Anderson bridge with a neat diagram. Explain its usefulness for measurement of self-inductance. Draw the phasor diagram when the bridge is at balance.
- Derive the expression for deflection torque of PMMC instruments.
- A Crompton's potentiometer consists of a resistance dial having 15 steps of 10 ohm each and a series connected slide wire of 10 Ω which is divided into 100 divisions. If the working current of potentiometer is 10 mA and each division of slide wire can be read accurately upto 1/5 th of its span, calculate the resolution of the potentiometer in volts.
- A dynamometer wattmeter reading correctly on D.C is used to measure power in circuit of resistance of 2 Ω and inductance of 0.25 H. The supply is 200 V at 50 Hz and the pressure coil circuit of wattmeter has a resistance of 1000 Ω and inductance of 5.6 mH. Calculate the actual reading of the wattmeter. Neglect the impedance of the current coil circuit. Assume the pressure coil is connected on load side of the instrument.
- What is power factor? Explain the construction and operation of crossed coil power factor meter.
- Explain the working of true rms responding voltmeter with suitable block diagram.
- Explain in detail the operation of digital storage oscilloscope with suitable block diagram.
- What are the functions of time base in frequency counter?

Part-III

Only Long Answer Type Questions (Answer Any Two out of Four)

- Q3** a) Explain the constructional difference between Wheatstone and Kelvin double bridge. **(16)**
b) Derive the equation of balance for the Kelvin Double bridge
c) The ratio arms of Kelvin bridge are 200Ω each. The Galvanometer has an internal resistance of 500Ω & a current sensitivity of $200\text{mm}/\mu\text{A}$. The unknown resistance $R_x=0.1002\Omega$ and standard resistance is set at 0.1Ω . A DC current of 10A is passed through the standard & the unknown from a 2.2V battery in series with a rheostat. The resistance of the yoke is neglected. Calculate i) the deflection of the galvanometer .ii) the resistance unbalance required to produce a galvanometer deflection of 1mm ?
- Q4** a) Describe the principle of working of a moving iron instrument. How are they classified? **(16)**
b) Show that this type of instrument can be used for both D.C and A.C measurements.
c) Illustrate the errors involved in both types of measurement.
- Q5** a) How the principle of operation of A.C. potentiometers differs from D.C. potentiometers and what are the factors that must be considered for operation of A.C. potentiometers? **(16)**
b) Explain about standardization and classification of A.C. potentiometers.
c) Compare the operation of Drysdale-Tinsley & Gall-Tinsley Potentiometer.
- Q6** a) What is the use of Q-meter? Describe basic Q-meter circuit. **(16)**
b) What are the methods used for connecting unknown components to the test terminal of Q-meter? Explain all the methods in details.

Registration No :

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Total Number of Pages : 03

B.Tech.

PEN2B101

2nd Semester Regular / Back Examination 2017-18

ELECTRICAL & ELECTRONICS ENGINEERING

BRANCH : AEIE, AERO, AUTO, BIOMED, BIOTECH,

CHEM, CIVIL, CSE, ECE, EEE, EIE, ELECTRICAL, ENV, ETC, FAT, IEE, IT, MANUFAC,

MANUTECH, MECH, METTA, MINERAL, MINING, MME, PE, PLASTIC, PT, TEXTILE

Time : 3 Hours

Max Marks : 100

Q.CODE : C705

Answer Part-A which is compulsory and any four from Part-B.

The figures in the right hand margin indicate marks.

Answer all parts of a question at a place.

Part – A (Answer all the questions)

Q1 Answer the following questions: multiple choice and fill in the blanks: (2 x 10)

- a) The resistance of a copper wire is R ohm. This wire is stretched to its double length. The new resistance is,
(i) $R\Omega$ (ii) $4R\Omega$ (iii) $2R\Omega$ (iv) $R/2\Omega$
- b) The average power in purely inductive circuit for one complete cycle is
(i) $E_{rms}I_{rms}$ (ii) zero (iii) $E_{max}I_{max}$ (iv) $(E_{max}I_{ma})/2$
- c) Binary representation of the decimal number 25 is
(i) 10001 (ii) 11001 (iii) 11101 (iv) 10110
- d) The current gain of BJT in common base is
(i) α (ii) β (iii) γ (iv) none of these.
- e) The rms value of 200 V DC supply is _____.
- f) The ripple factor of half wave rectifier is _____ and for full wave rectifier is _____
- g) The mobility of electrons in a materials is expressed in unit of _____
- h) A three Phase balanced delta connected load is connected to symmetrical three phase 440 V balanced supply. The current in each phase is 15 A and leads 60 degree ahead of the corresponding phase voltage. Then line current will be _____
- i) A two pole DC generator running at 1500 rpm has 40 conductors. The flux per pole is 1mWb. The induced emf if the armature winding is _____ for lap connected and _____ for wave connected.
- j) According to 1oolean law: $A + 1 =$ _____ and $A + 0 =$ _____

Q2 Answer the following questions: Short answer type: (2 x 10)

- a) Three resistor of resistance ($R_1=10\Omega$, $R_2=5\Omega$ and $R_3=3\Omega$) are connected in star network, convert it into delta network and find out its equivalent delta resistance .
- b) Define Unilateral and bilateral elements.
- c) Convert following number into decimal number (i) 34.85_8 (ii) 110.10101_2
- d) Draw the circuit diagram of half wave rectifier.
- e) State De morgen's theorem.
- f) Calculate the time taken by a capacitor of $1\mu F$ in series with a $1M\Omega$ resistance to be charged up to 80% of the final value.
- g) What is apparent power, active power and reactive power?
- h) What is P and N type semiconductor?

- i) A zener diode acts as a voltage regulator. Explain the meaning of the statement.
- j) What is the working principle of DC machines?

Part – B (Answer any four questions)

- Q3 a)** Find the current through $40\ \Omega$ resistor using superposition theorem for the circuit shown in figure 1. **(5)**

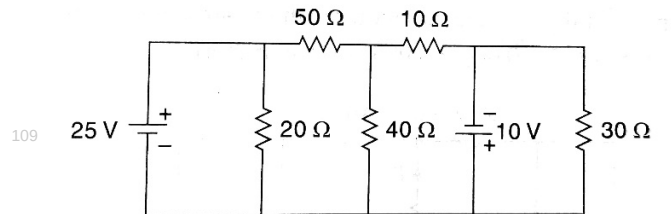


Figure 1

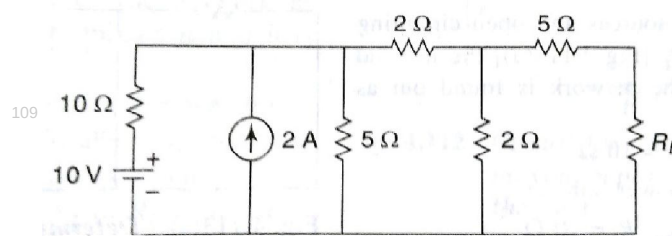


Figure 2

- b) State the maximum power transfer theorem and obtain the maximum power transferred to R_L in the circuit shown in Figure 2. And also find the value of R_L **(10)**
- Q4 a)** Explain the principle of operation of a transformer in detail and Derive the Emf equation of single phase transformer. **(5)**
- b) Draw the phase voltage and line voltage phasor diagram for 3-phase star connected balanced system. A 3-phase 230 V load has power factor of 0.7. Two wattmeter are connected to measure the power which shows the input to be 10 kW. Find the readings of each wattmeter. **(10)**
- Q5 a)** Explain the operation of a full wave bridge rectifier with relevant waveforms. **(5)**
- b) Explain the V-I characteristics of a P-N junction diode when it is connected in forward bias and reverse bias. A PN junction diode gives a current of 50 mA at a room temperature of 20 degree C when the forward bias voltage is 200mV. Determine (a) the saturation current with a negative bias (b) the diode current when room temperature is 30 degree C , and (c) diode current at a forward bias voltage. **(10)**
- Q6 a)** Explain the full adder circuit with its expression and truth table. **(5)**
- b) List out all the basic logic gates and universal gates with its logic symbols and truth table. And generate AND function, OR function and NOT function using any one Universal gate. **(10)**

Q7 a) A series circuit has $R = 5 \Omega$, $L = 13 \text{ mH}$ and $C = 140 \mu\text{F}$ and is supplied with 230 V , 50 Hz single phase. Find (i) Impedance (ii) current (iii) power (iv) power factor of the circuit. **(8)**

b) An iron ring made up of three parts, $l_1 = 12 \text{ cm}$, $a_1 = 6 \text{ cm}^2$, $l_2 = 10 \text{ cm}$, $a_2 = 5 \text{ cm}^2$, $l_3 = 8 \text{ cm}$ and $a_3 = 4 \text{ cm}^2$. It is surrounded by a coil of 200 turns. Determine the exciting current required to create a flux of 0.5 mwb in the iron ring. [Given $\mu_1 = 2670$, $\mu_2 = 1055$, $\mu_3 = 680$] **(7)**

Q8 a) Simplify the function $Y = (A+B)(\bar{A} + C)(B+C)$ and design the circuit for simplified function using basic logic gates. **(8)**

b) Determine (i) $I_{C(\text{sat})}$ (ii) I_C (iii) V_C (iv) V_E and (v) V_{CE} for the circuit shown in figure 3. **(7)**

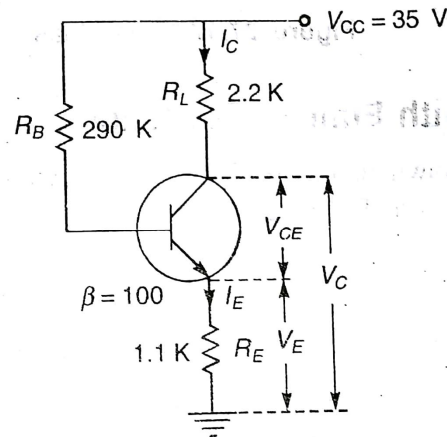


Figure 3

Q9 Write a short note on any THREE : (5 x 3)

- Magnetic material and B-H curve
- Different methods of transistor biasing.
- Generation and distribution of AC Power
- Transients in RL circuit with DC excitation
- circuit elements and their characteristics

Registration No :

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Total Number of Pages : 02

B.Tech.
PET4I103

4th Semester Regular / Back Examination 2017-18
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH : ECE, ETC
Time : 3 Hours
Max Marks : 100
Q.CODE : C1013

Answer Part-A which is compulsory and any four from Part-B.

The figures in the right hand margin indicate marks.

Answer all parts of a question at a place.

Part – A (Answer all the questions)

Q1 Answer the following questions :

(2 x 10)

- a) A reading is recorded as 23.90°C. The reading has number of significant figures _____.
a. 4 b. 3 c. 2 d. 1
- b) A 0-10A ammeter has a guaranteed accuracy of 1% of full scale deflection. The limiting error while reading 2.5 A is:
a. 1% b. 4% c. 2% d. none of the above
- c) Find multiplying power of a shunt of 200Ω resistance used with a galvanometer of 1000Ω resistance.
a. 4 b. 6 c. 5 d. 4.5
- d) In a 3-phase power measurement by using 2-wattmeter method, both the meters give equal readings, find the power factor of the load.
a. 0.7 b. 0 c. 0.5 d. 1
- e) Standardization of potentiometer is done in order that, they become
a. accurate and precise b. accurate and direct reading
c. Precise d. accurate
- f) A Wheatstone bridge can not be used for precision measurements, because errors are introduced into on account of,
a. Resistance of connecting leads b. Thermo electric emfs
c. Contact resistances d. All the above
- g) A current transformer has a rating of 100/5A. Its magnetizing and loss component of the exciting current are 1A and 0.6A respectively and secondary winding burden is purely resistive, its transformation ratio at rated current is
a. 20.12 b. 20.2 c. 200.2 d. None of the above
- h) Electronic voltmeters which use rectifiers employ negative feedback. This is done
a. To increase the overall gain b. To improve stability
c. To overcome non-linearity of diodes d. None of the above
- i) Post acceleration is needed in a CRO, if the frequency of the signal is
a. <1Mhz b. >1Mhz c. >10Mhz d. >10kHz
- j) Hay's bridge is used to measure
a. Resistance b. Inductance c. Capacitance d. Frequency

Q2 Answer the following questions: Short answer type: (2 x 10)

- a) What is dead time and dead zone?
- b) State the types of error in measurement system.
- c) What is logarithmic decrement?
- d) Discuss the effect of temperature changes in voltmeter.
- e) Discuss the errors found in current transformers.
- f) What are the advantages and disadvantages in PMMC instruments?
- g) What are the considerations in selecting an analog voltmeter?
- h) Write some of the applications of CRO.
- i) What is the effect of beam transit time on deflection sensitivity?
- j) What is total harmonic distortion?

Part – B (Answer any four questions)

- Q3 a)** Derive the equation for balance in case of Maxwell's inductance bridge. Draw the phasor diagram for balanced condition. **(10)**
- b)** The ratio arms of a kelvin bridge are 100Ω each. The galvanometer has an internal resistance of 500Ω and a current sensitivity of $200 \text{ mm}/\mu\text{A}$. The unknown resistance is 0.1002 and the standard resistance is set at 0.1000Ω . A d.c. current of 10 A is passed through the standard and unknown resistance from a 2.2V battery in series with a rheostat. Calculate the deflection of the galvanometer, neglect the resistance of the link. Find also the resistance unbalanced to produce a deflection of 1mm . **(5)**
- Q4 a)** Describe the working of Schering bridge. Derive the equations for capacitance and dissipation factor, draw the phasor diagram under balanced condition. **(10)**
- b)** The coil of a ballistic galvanometer has a resistance of 150Ω and an undamped period of 7.5s . A steady emf of 3.5 mV produces a deflection of 210mm . Determine the quantity of electricity discharged from a capacitor if the deflection produced is 750mm . The relative damping is 0.8 . **(5)**
- Q5 a)** Derive the torque equation for an electro-dynamometer type wattmeter. **(10)**
- b)** Draw the block diagram of a ramp type DVM. **(5)**
- Q6 a)** Explain theory and principles of operation of Gall-Tinsley co-ordinate type AC potentiometer. **(10)**
- b)** Explain loss of charge method. **(5)**
- Q7 a)** Discuss advantage and disadvantage of analogue and digital type of oscilloscope. **(10)**
- b)** Explain the working of a differential amplifier using two FETs. **(5)**
- Q8 a)** Discuss the basic circuit of a spectrum analyser. **(10)**
- b)** Describe the circuit diagram and operation of a true rms reading voltmeter using thermocouples. **(5)**
- Q9 a)** Describe the constructional details and principle of operation of a d'Arsonval galvanometer. Derive the expression for steady state deflection. **(10)**
- b)** Write short notes on Q-meter. **(5)**

Registration No:

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Total Number of Pages: 03

B.TECH
PET4I103

4th Semester Regular Examination 2016-17
ELECTRICAL & ELECTRONICS MEASUREMENT
BRANCH(S): ECE, ETC

Time: 3 Hours

Max Marks: 100

Q.CODE: Z553

Answer Part-A which is compulsory and any four from Part-B.
The figures in the right hand margin indicate marks.

Part – A (Answer all the questions)

Q1 Answer the following questions: *multiple type or dash fill up type* (2 x 10)

- a) The insulation resistance of a transformer winding can be easily measured with (a) Wheatstone bridge (b) megger (c) Kelvin bridge (d) voltmeter
- b) The internal resistance of an ammeter should be (a) Very small (b) Medium (c) High (d) Infinity
- c) The following error does not result in moving iron instrument with both DC and AC: (a) Stray magnetic field error (b) Hysteresis error (c) Eddy current error (d) Temperature error
- d) Damping of the ballistic galvanometer is kept very small (a) To make the system oscillating (b) In order to get first deflection small (c) Make the system critically damped (d) In order to get first deflection large.
- e) X and Y plates of a CRO are connected to unequal voltages of equal frequency with phase shift of 90° . The Lissajous figure on the screen will be (a) Circle (b) Straight line (c) Ellipse (d) Figure of eight.
- f) Which bridge is used for measuring audio and H.F. oscillation? (a) Wein bridge (b) Maxwell bridge (c) Schering bridge (d) None of the above.
- g) Number of significant figure in 0.130450 is (a) 7 (b) 5 (c) 3 (d) 6
- h) High quality factor (Q) of an inductor can be measured by: (a) Hay's bridge (b) Anderson bridge (c) Wien bridge (d) Schering bridge
- i) The rating of a battery is given by: (a) KW (b) KVA (c) Ampere-hours (d) VARh
- j) To determine the voltage polarity across a resistor we have to know: (a) The battery voltage (b) The value of the resistor (c) The temperature of the resistor (d) The direction of the current

Q2 Answer the following questions: *Short answer type* (2 x 10)

- a) Define logarithmic decrement and what its unit is.
- b) If a voltmeter of range 0-250V has an error of 1%, find the % error when the voltmeter reads 125 V.

- c) In a Drysdale polar potentiometer, what is the use of a variable capacitance?
- d) Why Kelvin's double bridge is superior to Wheatstone bridge for the purpose of low resistance measurement?
- e) What is meant by Q-factor of a coil?
- f) Can the CT be energized while the secondary winding is kept open? Explain.
- g) A $3\frac{1}{2}$ digit voltmeter is used for voltage measurement. Find its resolution.
- h) Draw the block diagram of a true RMS voltage.
- i) Why an ordinary electro dynamometer type wattmeter is not suitable for measurement of power in a low power factor circuit?
- j) What is a Lissajous pattern?

Part – B (Answer any four questions)

- Q3 a)** Explain measurement of earth resistance by fall of potential method with suitable diagrams. **(10)**
- b)** In the measurement of low resistance by potentiometer, the voltage drop across a resistor under test and across a 0.1Ω standard resistor were found to be 0.835V and 0.984V respectively. The resistance of the standard resistor at the test temperature is 0.10025Ω . upon setting the potentiometer dials to zero and interrupting the current passing through the unknown resistance, the thermal emf of the latter caused a galvanometer deflection equivalent to $25\mu\text{V}$, the direction the emf being opposite to that produced by an increase of the potentiometer reading during voltage measurement. Determine the value of the resistor under test? **(5)**
- Q4 a)** Explain the operation of Schering bridge with diagram. **(10)**
- b)** An ac bridge ABCD has the following four arms taken in sequence: **(5)**
- Arm AB: a capacitance with series loss effect resistance R in series with the primary of a mutual inductance M.
 - Arm BC: resistance R_2
 - Arm CD: resistance R_3
 - Arm DA: lossless capacitance C_4 .
- The primary and the mutual inductance has self-inductance L and negligible resistance. The secondary of the mutual inductance is connected in series with detector across BD. The source of angular frequency ω is connected across AC. Determine the capacitance C and associated loss angle if the bridge is balanced with $\omega=10^3$, $M=0.0015\text{H}$, $R_2=R_3=10^4\Omega$, $L=0.0045\text{H}$ and $C_4=0.15\mu\text{F}$.

- Q5 a)** A current transformer with a bar has 300 turns in its secondary winding. The resistance and the reactance of the secondary winding are 1.5Ω and 1.0Ω respectively. With 5A flowing in the secondary winding, the magnetizing mmf is 100 ampere turns and the iron loss is 1.2 watts. Determine the ratio and the phase angle errors. **(10)**
- b)** Derive the expression for ratio error in a potential transformer. **(5)**
- Q6 a)** Describe the power measurement by 2-wattmeter method in 3-phase system with phasor diagram. **(10)**
- b)** What are the lag adjustment devices used in energy meters and explain them **(5)**
- Q7 a)** Describe the construction of a Ballistic galvanometer. Explain its theory of operation. Compare this galvanometer with the ordinary dc galvanometer. How do you determine the logarithmic decrement? **(10)**
- b)** Describe the working of a Q-meter with circuit diagram. **(5)**
- Q8 a)** Explain the operation of a basic spectrum analyzer. How the spectra of a amplitude modulated signal is displayed? **(10)**
- b)** Describe the operation of true RMS voltmeter. **(5)**
- Q9 a)** Explain the phase and frequency measurement of ac quantity by CRO. **(10)**
- b)** A 3 phase 3-wire, 415V system supplies a balanced load of 20A at a power factor of 0.8 lagging. The current coil of wattmeter 1 is in phase 'R' and of wattmeter 2 in phase 'B'. calculate I) the reading on 1 when it's voltage coil is across 'R and Y', II) the reading on 2 when it's voltage coil is across 'B and Y' **(5)**

Registration no:

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Total Number of Pages: 02

B.Tech
PCEE4204

3rd Semester Back Examination 2016-17
ELECTRICAL AND ELECTRONICS MEASUREMENT
BRANCH(S): AEIE, ECE, EIE, ETC
Time: 3 Hours
Max Marks: 70
Q.CODE: Y584

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions: (2 x 10)

- Give at least two most common methods for measurement of low resistance.
- List any four Static characteristics of a measuring system.
- What is exact difference between accuracy and Precision?
- Mention the two modes of operation in dual trace oscilloscope.
- Give the functions of an attenuator in a signal generator.
- What is the important of gate time in frequency counter?
- Give any two applications of micro processor based measurement.
- Mention the term used to specify the characteristics of an instrumentation amplifier.
- Differentiate between sensors and transducers.
- What are the sources of errors in D.C voltage measurement?

Q2 With help of neat sketch, explain the construction and working of an electrodyname meter power factor meter. (2+8)

Q3 a) Design a multi range d.c millimeter with a basic meter having a resistance of 75ohms and full scale deflection for a current of 2mA. The required ranges are 0-10mA, 0-50mA and 0-100Ma. (6)

b) Explain the Kelvin's Double bridge and obtain the balance condition. (4)

Q4 a) Briefly explain the calibration and adjustments of a single phase induction type energy meter. (5)

b) Explain the operation of LVDT, with help of a diagram. (5)

Q5 a) How the frequency is converted to an analog signal? Explain. (5)

b) Explain any one bridge circuit for measurement of Inductance. (5)

Q6 a) Briefly explain the basic elements of a digital acquisition system. **(5)**

b) For 20A, 230V energy meter, the revolution per Kilowatt-hour is 480. If upon test at full load unit power factor the disc makes 40 revolution in 66 seconds, calculate the error in the energy meter. **(5)**

Q7 With a neat schematic, explain the operation of a dual slope analog to digital conversion. **(10)**

Q8 Write short answer on any TWO: (5 x 2)

a) Photo Voltaic Cell

b) X-Y recorder

c) Megger

d) Ramp type DVM

Registration No:

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Total Number of Pages: 02

B.TECH
PEE3I104

3rd Semester Regular Examination 2016-17
ELECTRICAL AND ELECTRONICS MEASUREMENT

BRANCH: EE

Time: 3 Hours

Max Marks: 100

Q.CODE:Y581

Answer Part-A which is compulsory and any four from Part-B.

The figures in the right hand margin indicate marks.

Part – A (Answer all the questions)

Q1 Answer the following questions: *multiple type or dash fill up type* **(2 x 10)**

- An ammeter of 0-25 A range has a guaranteed accuracy of 1% of full scale reading. The current measured is 5 A. The limiting error is -----
- In 3 phase power measurement by two wattmeter method, the reading of one wattmeter is zero. The power factor of load is -----
- A single phase energy meter has the rating 1200 revolutions/ kWh. If a 500 W electric gadget is used for 4 hours, the energy meter will make -----
- In which of the transformer is the secondary nearly short circuited under normal operating conditions?
A. CT B. PT C. Distribution transformer D. Power transformer
- A moving coil instrument has a resistance of 0.6Ω and full scale deflection at 0.1 A. To convert it into an ammeter of 0-15 A range, the resistance of shunt should be -----
- In a ballistic galvanometer, a charge of $100 \mu\text{C}$ gives a first swing of 25° . The charge required to cause first swing of 50° is -----
- Two resistors $R_1 = 36 \Omega \pm 5\%$ and $R_2 = 75 \Omega \pm 5\%$ are connected in series. The total resistance is -----
- A digital voltmeter has a read out range from 0 to 999 counts. If the full scale reading is 9.999 V, the resolution is -----
- The coil of a moving coil meter has 100 turns, is 40 mm long and 30 mm wide. The control torque is $240 \times 10^{-6} \text{ N-m}$ on full scale. If $B = 1 \text{ Wb/m}^2$, range of meter is -----
- A standard cell of 1.0185 V is used with a slide wire potentiometer. The balance was obtained at 60 cm. When an unknown emf was connected, the balance was obtained at 82 cm. The magnitude of unknown emf is -----

Q2 Answer the following questions: *Short answer type* **(2 x 10)**

- Explain felici's method for measurement of mutual inductance.
- Define different types of standards of measurement.
- Define constants of galvanometer.
- Discuss the causes of errors in single phase energy meter.
- What is loss of charge method.
- Write galvanometer motion equation and state conditions for different types of damping behavior.

- g) Define phantom loading.
- h) Discuss the effect of secondary current on potential transformer ratio.
- i) Electrodynamometer instruments are called transfer instruments, Explain.
- j) Draw the block diagram of a digital storage oscilloscope.

Part – B (Answer any four questions)

- Q3 a)** What are the different problems associated with measurement of low resistances and how these are eliminated using Kelvin' Double Bridge? Draw the circuit of a Kelvin' Double Bridge and derive the condition for balance. **(10)**
- b)** The four arms of a wheatstone bridge are as follows: **(5)**
 $AB=100\Omega$; $BC=1000\Omega$, $CD=4000\Omega$, $DA=400\Omega$. The galvanometer has a resistance of 100Ω and, a sensitivity of $100\text{mm}/\mu\text{A}$ and is connected across AC. A source of 4 V d.c. is connected across BD. Calculate the current through the galvanometer and its deflection if the resistance of arm DA is changed from 400Ω to 401Ω .
- Q4 a)** Derive the equations for balance in case of Maxwell's inductance capacitance bridge (for unknown inductance) and De Sauty's bridge (for unknown capacitance). Draw the phasor diagrams for balance conditions. **(10)**
- b)** Explain the function and working of Wagner Earth Devices. **(5)**
- Q5 a)** Derive expression of the deflecting torque of a moving iron instrument. The inductance of a moving iron instrument is given by:
 $L = (0.01 + A\theta)^2\text{mH}$, where θ is the deflection from zero position in degree. The angular deflections of the instrument corresponding to 1.5 and 2 A are respectively 90° and 120° . Find the value of A. **(10)**
- b)** Derive the expression for external resistance required for critical damping of a galvanometer. **(5)**
- Q6 a)** Describe the working principle for induction type energy meter and derive the torque expression. A 230 V, single phase, watt hour meter has a constant load of 4 A passing through it for 6 hours at unity power factor. If the meter disc makes 2208 revolutions during this period what is the meter constant in revolutions per kWh? Calculate the power factor of the load if the number of revolutions made by the meter are 1472 when operating at 230 V and 5 A for 4 hours. **(10)**
- b)** Derive the torque equation for a dynamometer type wattmeter. Comment upon the shape of scale if spring control is used. **(5)**
- Q7 a)** Describe the construction and working of a polar type potentiometer. **(10)**
- b)** Write short notes on flux meter. **(5)**
- Q8 a)** Derive the expressions for ratio and phase angle errors for current transformer and discuss the effect of secondary burden on it. **(10)**
- b)** Write short notes on ramp type DVM. **(5)**
- Q9 a)** Draw block diagram of an oscilloscope and explain its operation. Explain the methods of measurement of frequency, phase angle and time delay using oscilloscope. **(10)**
- b)** Write short notes on rectifier type AC voltmeters **(5)**

Registration No:

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Total Number of Pages: 02

B.TECH
PCEE4204

3rd Semester Regular / Back Examination 2015-16
ELECTRICAL AND ELECTRONICS MEASUREMENT

BRANCH: AEIE,EC,EIE,ETC,IEE

Time: 3 Hours

Max Marks: 70

Q.CODE: T555

Answer Question No.1 which is compulsory and any five from the rest.

The figures in the right hand margin indicate marks.

- Q1** Answer the following questions: **(2 x 10)**
- a) What are the dynamic characteristics of an instrument?
 - b) A regular Wheatstone bridge is used to measure a mega ohm resistance. The bridge has ratio arms of 9988 and 10. The variable arm has a maximum resistance of 9999. What is the maximum resistance that can be measured by this arrangement.
 - c) Define logarithmic decrement and what its unit is.
 - d) In a dynamometer type instrument, the mutual inductance varies with deflection as $-6\cos(30+\Theta)$ mH. Find the deflecting torque produced by a D.C current of 50mA corresponding to a deflection of 60.
 - e) In a Drysdale polar potentiometer, what is the use of a variable capacitance?
 - f) What are the disadvantages of moving Iron type power factor meter?
 - g) In order to minimize the errors in a current transformer, the magnetizing and loss component in an instrument transformer must be kept low.
 - h) Draw the block diagram of a true RMS voltage.
 - i) Find the resolution of a $3\frac{1}{2}$ digit digital voltmeter used for measuring 0 to 1 volt.
 - j) How does an amplifier introduce harmonic distortion of a signal in the output?
- Q2** a) Explain the operation and working of Wagner Earthing device with neat circuit diagram. **(5)**
- b) A balanced Hay's bridge has $R_2=R_3=1000\Omega$, $R_4=8120\Omega$, $C_4=981\text{pF}$ and the frequency is 4000Hz. Calculate L_1 & R_1 . Also find out the Q-factor of the coil. **(5)**
- Q3** a) What is sensitivity and what are the different ways of defining it in galvanometers. Write the expressions for all. **(5)**
- b) A single range potentiometer has a 18 step dial switch, where each step represents 0.1Volt. The dial resistors are 10Ω . The slide wire of the potentiometer is circular and has 11 turns and a resistance of 11Ω each. The slide wire has 100 divisions and interpolation can be done to one fourth of division. The working battery has voltage of 6Volt and negligible internal resistance. Calculate (a) the measuring range (b) resolution (c) working current (d) setting of the rheostat **(5)**

- Q4** What are the errors in electro-dynamometer type Wattmeter's? Explain them **(10)**
- Q5** a) The ratio error of a given 1000/5 A CT is zero when feeding 5VA, unity power factor burden at rated current. Estimate the iron loss of the transformer at this operating condition if the secondary has 198 turns and a winding resistance of 0.02Ω . Neglect leakage reactance. **(5)**
- b) A wattmeter has current coil resistance of 0.03Ω and potential coil resistance of $6k\Omega$. Calculate the percentage error if the wattmeter is so connected that: (i) the current coil is on the load side. (ii) The potential coil is on the load side. The load takes 20A at a voltage 220V and 0.6 power factor in each case. **(5)**
- Q6** a) Describe the working of a Q-meter with circuit diagram. **(5)**
- b) Describe capacitance measurement by Owen's bridge with circuit diagram and expressions. **(5)**
- Q7** a) Explain the phase and frequency measurement of ac quantity by CRO. **(5)**
- b) Describe the working of a wave analyzer **(5)**
- Q8** Write short notes on any two: **(5 x 2)**
- a) Explain measurement of charge by Ballistic Galvanometer.
- b) Describe the working of a power factor meter.
- c) Measurement of mutual inductance by Felici's method.
- d) Derive the expressions for Deflecting torque in PMMC type ammeter and voltmeter meter

Registration no:

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Total Number of Pages: 2

B.Tech
PCEE4204

4TH Semester Regular / Back Examination 2015-16
ELECTRICAL AND ELECTRONICS MEASUREMENT

BRANCH: EEE, ELECTRICAL

Time: 3 Hours

Max Marks: 70

Q.CODE: W481

Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right hand margin indicate marks.

Q1 Answer the following questions: **(2 x 10)**

- The guaranteed accuracy of a flow meter working on thermal principle is ± 3 percent of full scale reading of $2.5 \times 10^{-6} \text{ m}^3/\text{s}$. If the flow measured by this meter is $1.25 \times 10^{-6} \text{ m}^3/\text{s}$. Calculate the limiting error in percent.
- Define the term Backlash as applied to the measurement.
- Write the functions of Triggering circuit in an oscilloscope.
- How to reduce the error in a Current Transformer?
- Name four intrinsic constants of a galvanometer.
- What is the function of attenuators in CROs?
- Why electro-dynamometer type instruments are called as "Transfer Instrument"?
- How to Standardize a DC Potentiometer?
- List various methods used for the measurement of low, medium and high resistances.
- How the construction of a D'Arsonval type galvanometer is different from Ballistic galvanometer?

Q2 a) Explain the working of Laboratory type Crompton's Potentiometer? **(5)**

- b)** The three resistors R_1 , R_2 and R_3 have the following ratings: **(5)**
 $R_1 = 25\Omega \pm 4\%$, $R_2 = 65\Omega \pm 4\%$, $R_3 = 45\Omega \pm 4\%$
Determine the following:
- Limiting value of the resultant resistance.
 - Percentage limiting error of series combination of resistance.

Q3 a) Derive an expression of the deflecting torque of a Moving Iron Instrument. **(5)**

- b)** Explain briefly, how inductance can be measured using Maxwell Bridge? **(5)**

Q4 Derive the expression for Transformation Ratio error and Phase angle error in a Current Transformer. **(10)**

- Q5** a) Describe the construction and working of a Ballistic galvanometer? (5)
b) What are creeping and phantom loading? Explain in brief. (5)
- Q6** a) Explain the function of focusing anode used in Cathode Ray Tube? (5)
b) Write in brief the function of Electron gun and Control Grid used in Cathode Ray Tube? (5)
- Q7** a) Describe the construction and operation of a Polar type AC Potentiometer. (7)
b) What is the difference between Polar type and Co-ordinate type Potentiometer? (3)
- Q8** Write short notes on any two: (5 x 2)
a) Errors in Electrodynamometer type of Instruments.
b) Causes of error in Current Transformer.
c) Digital frequency counter.
d) Dual ramp type digital voltmeter.